

Answer **all** questions in the spaces provided.

0 1 . 1 What is a gene?

[1 mark]

0 1 . 2 Describe how the production of messenger RNA (mRNA) in a eukaryote cell is different from the production of mRNA in a prokaryote cell.

[2 marks]

0 1 . 3 Scientists produced a short, single-stranded, artificial nucleic acid, called PNA. The PNA binds to a small section of DNA.

The scientists introduced PNA into cells and discovered that these cells produced less mRNA than cells that did not contain PNA.

Suggest how PNA affected the transcription of the section of DNA.

[2 marks]



0 1 . 4

Describe the role of ATP in the process of translation in protein synthesis.

[2 marks]

7**Turn over for the next question****Turn over ►**